DATS 6501 CAPSTONE · PRELIMINARY PRESENTATION

SynthAug-Bench

When does diffusion-generated data help image classification?

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Can generated images stand in for scarce real ones?

1

Labelled images are expensive

Many problems have only a handful of images per class.

2

Text-to-image models are free to sample

Stable Diffusion can produce unlimited images for any class name.

3

But do they beat cheap augmentation?

Most papers test one model, one data size and one baseline.

MAIN QUESTION

Under which data, model and task conditions does diffusion-based augmentation beat real-only training and conventional augmentation, and when does it hurt?

STAGE 1 FOCUS

- RQ2: do a CNN and a Transformer respond differently?
- RQ4: which settings make it worse than RandAugment?
- Does ImageNet pretraining change the answer?

Five stages, each one started only when the last has results

1

Core classification

CIFAR-100 · ResNet-50, ViT-B/16 · RandAugment, SD, SD + LoRA

2

Full CIFAR-100

ConvNeXt-T, Swin-T · Mixup, CutMix · textual inversion · ratios 2:1, 5:1 · 5 seeds

3

Other data regimes

Long-tail CIFAR-100-LT · fine-grained Flowers, CUB, Pets · medical HAM10000

4

Generators from scratch

StyleGAN2-ADA and a small DDPM · does image quality predict the gain?

5

Beyond classification

MS COCO 2017 detection and captioning · pretrained-feature hybrid

Stage 1: one comparison, repeated 300 times

5

Data budget

5 · 10 · 20 · 50 per class, and the full 45,000

×

2

Classifier

ResNet-50 (CNN)
ViT-B/16 (Transformer)

×

5

Condition

Real only · RandAugment · SD class prompts · SD + LoRA · SD + RandAugment

×

2

Initialization

ImageNet-pretrained
random

×

3

Training seed

0 · 1 · 2

= 300 runs

Synthetic images are added 1 : 1 to the real images of each budget and reduced to the 32 × 32 CIFAR resolution.

Metrics: top-1 accuracy, macro-F1, balanced accuracy and per-class recall on held-out real images.

The protocol was fixed before any comparison was made

1

Split manifest

50 validation images per class (5,000) taken from the training set. Nested budgets, split seed 42, label fingerprint checked on every run.

2

Pilots

Learning rate and step budget chosen per model and initialization, on validation, before any synthetic run.

3

Same steps for all

Every condition gets the same number of optimizer steps, so a gain is not just longer training. Best checkpoint picked on validation.

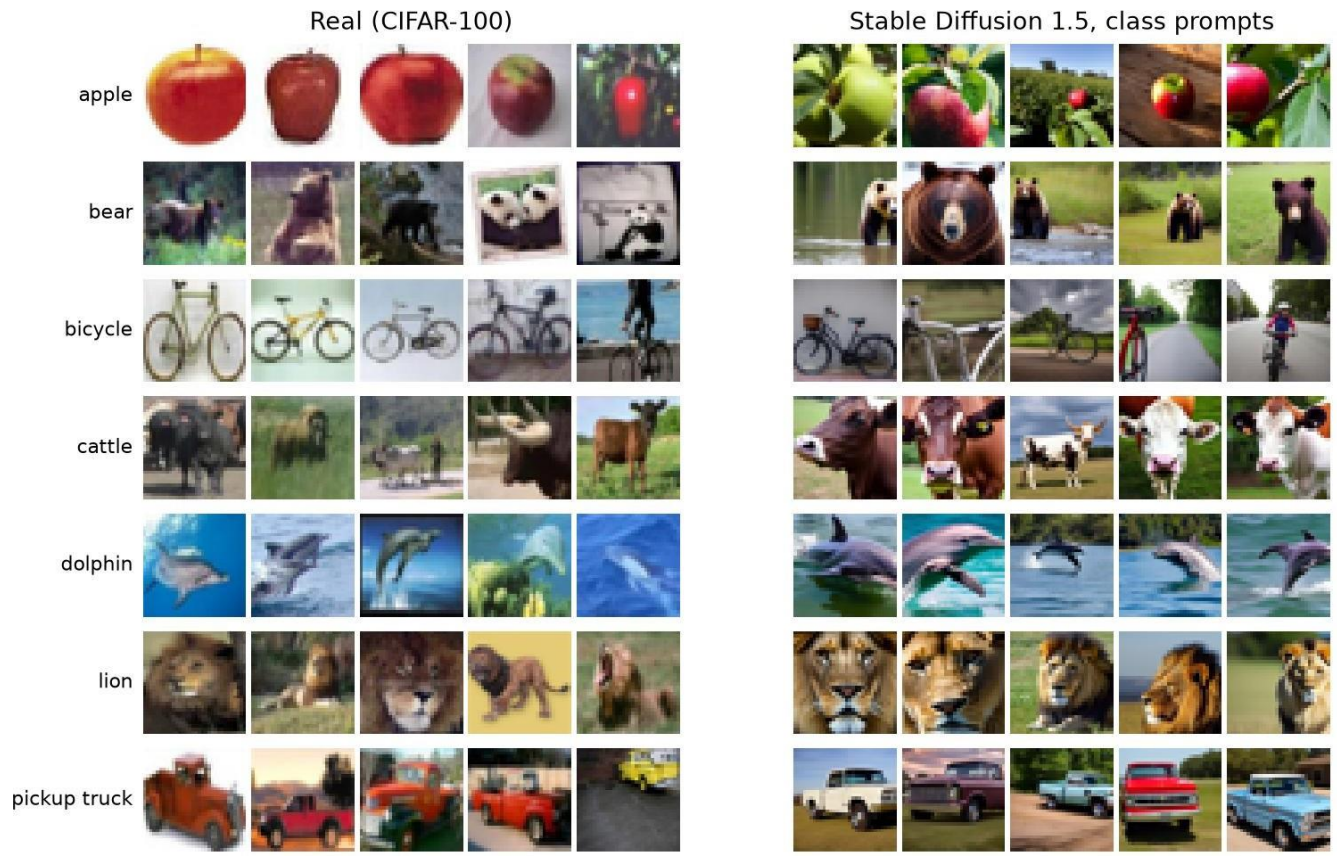
4

Paired seeds

Each seed is compared with the same seed of its control. A gain or loss is declared only when all three seeds agree.

Test set held out: the official 10,000 test images are scored once, after the protocol is frozen. All results today are validation results.

98,500 generated images from Stable Diffusion 1.5



Real CIFAR-100 (left) and Stable Diffusion class-prompt samples (right), shown at 32 × 32

SD class prompts

45,000 images · 8 prompt templates and a negative prompt · DPM-Solver++, 25 steps, guidance 7.5

SD + LoRA

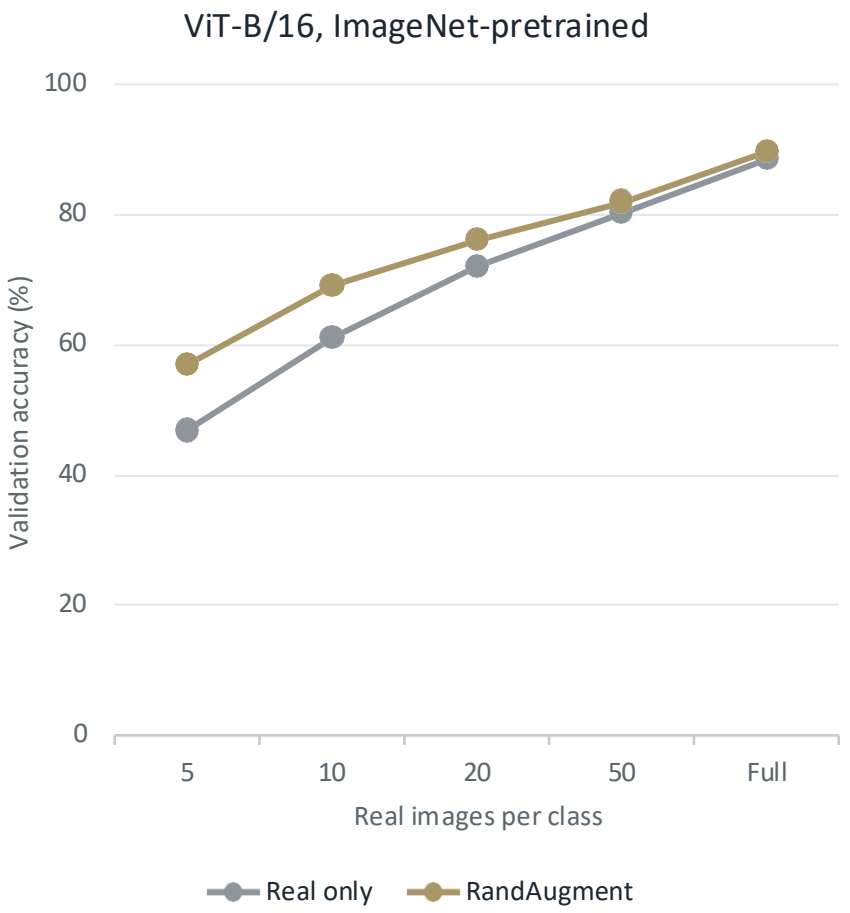
One rank-8 adapter per budget, trained only on that budget's real images · 53,500 images

Quality

FID and Improved Precision & Recall against an equal-size real sample, with a real-vs-real floor

Pretrained baselines: RandAugment helps at every budget





+10.1

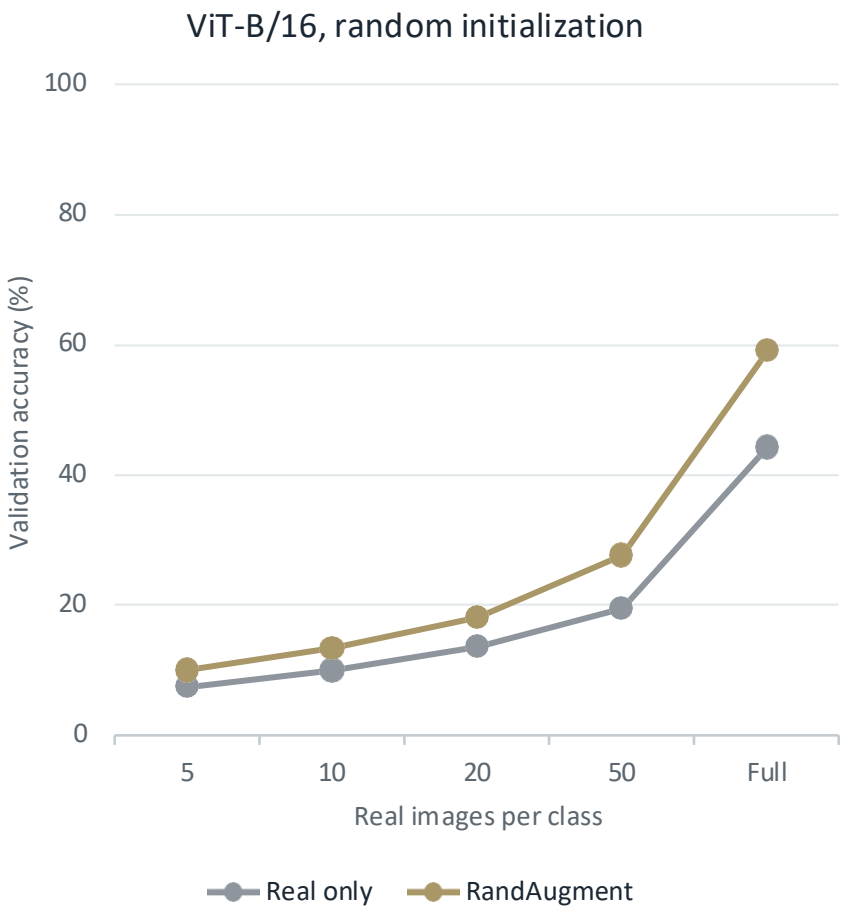
points for ViT-B/16 at 5 images per class (ResNet-50: +7.9)

~1

point left on the full training set. Gain in all 3 seeds at every budget

From scratch, accuracy collapses when data is scarce





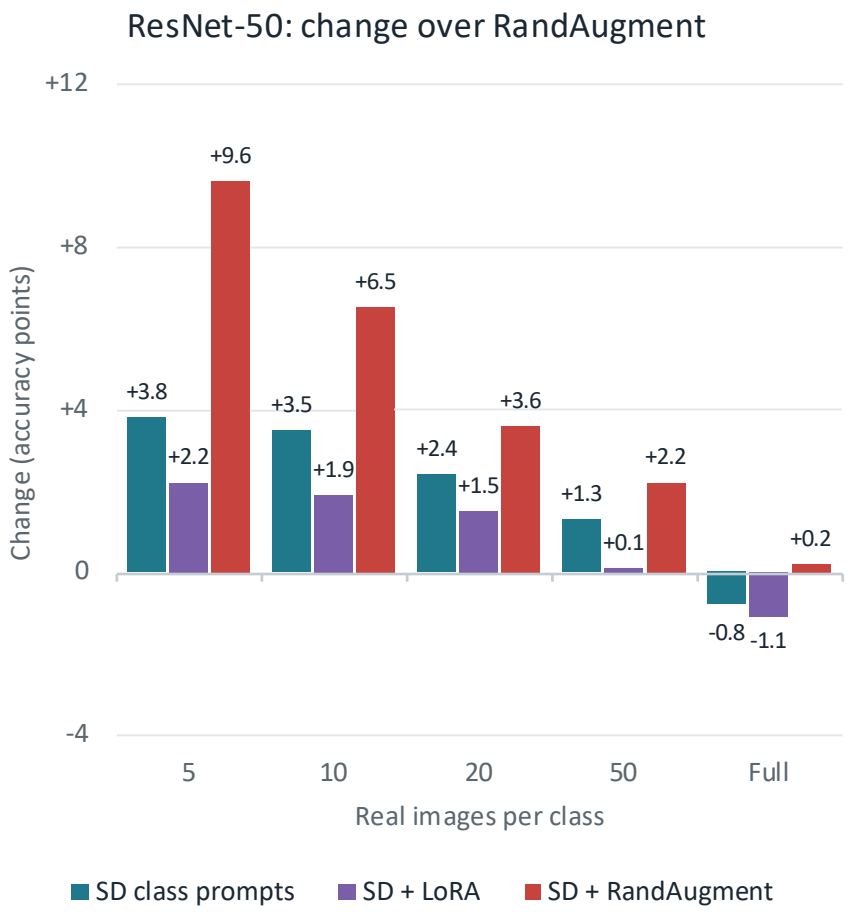
8.9%

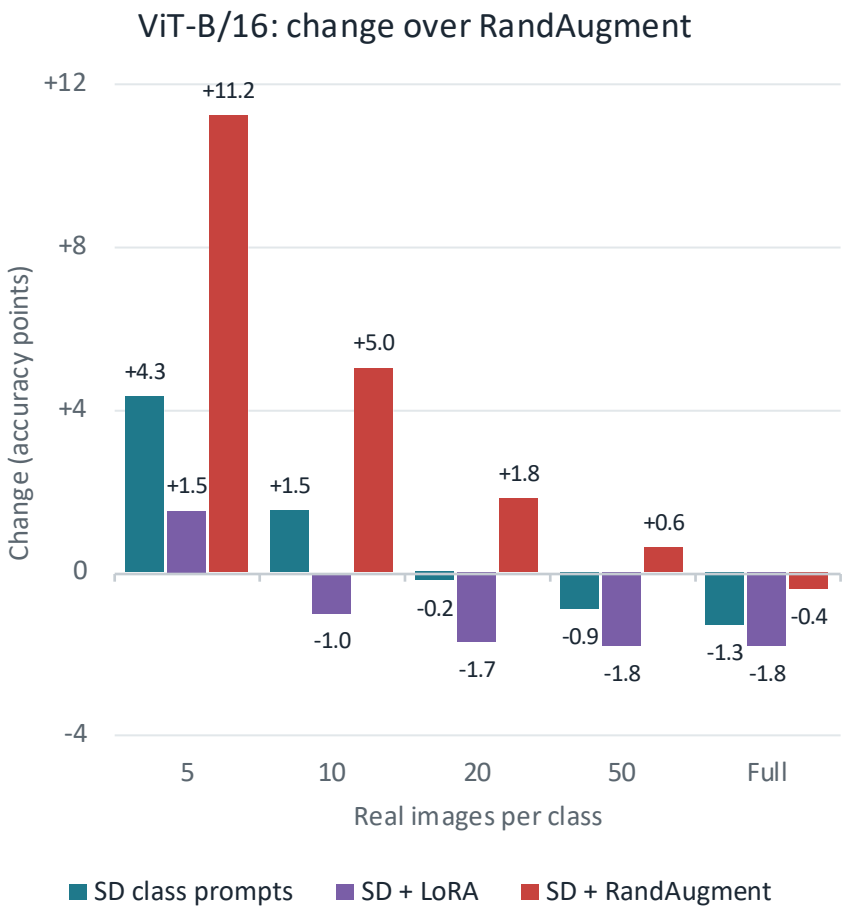
ResNet-50 at 5 images per class from scratch, against 35.2% pretrained

+15.0

points from RandAugment for ViT-B/16 on the full set: scratch models need it most

Pretrained: SD + RandAugment wins up to 50 per class





+11.2

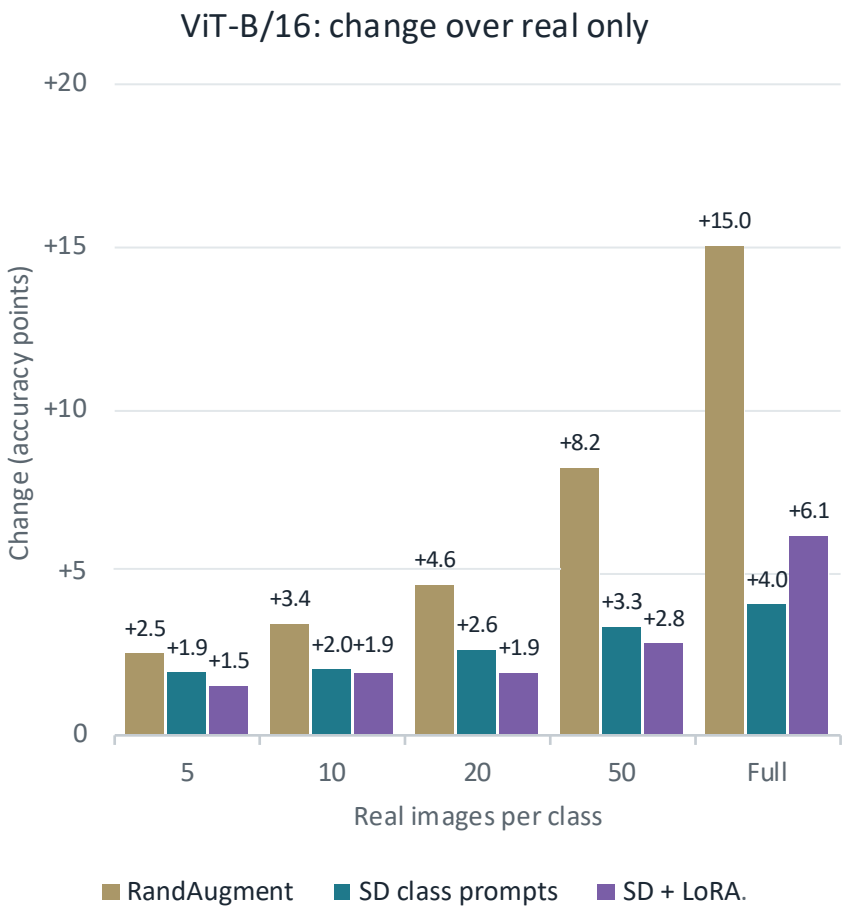
points for ViT-B/16 at 5 per class (ResNet-50: +9.6), in all 3 seeds

≈ 0

on the full set: synthetic images on their own fall behind RandAugment there

From scratch, RandAugment beats synthetic images





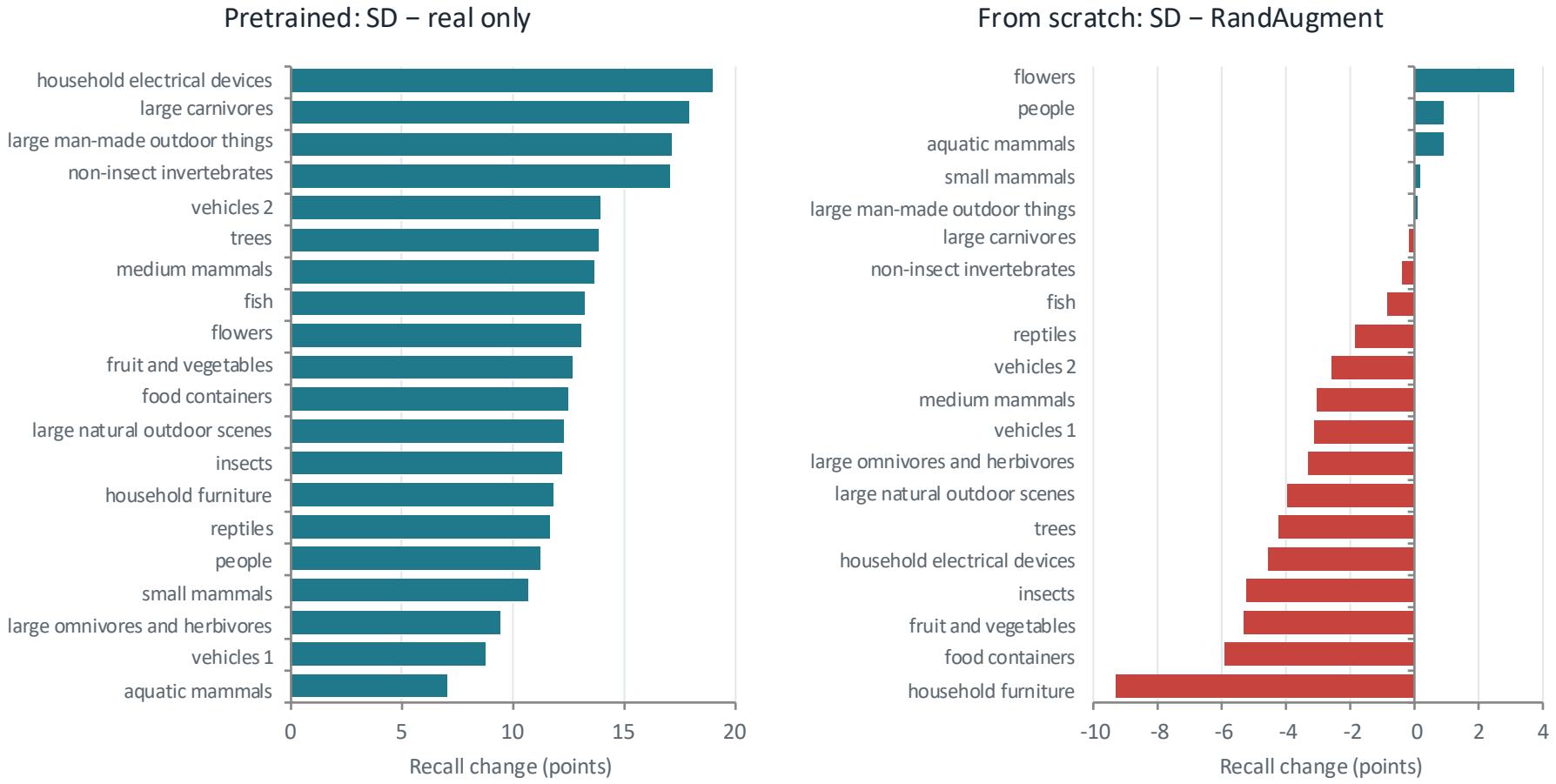
20 / 20

synthetic comparisons beat real only, in every seed

18 / 20

lose to RandAugment; the other 2 are inconclusive

Pretrained models gain in every superclass



20 / 20

superclasses gain from SD images with pretrained models

15 / 20

superclasses where SD trails RandAugment from scratch

Recall change at 5 real images per class, mean of both classifiers and 3 seeds; 250 validation images per superclass.

A closer generator did not give a better classifier

Per class	FID, SD	FID, LoRA	Real-vs-real floor	Recall, SD	Recall, LoRA
5	**103.1**	104.8	88.9	0.43	**0.50**
10	71.9	**71.2**	54.3	0.36	**0.42**
20	48.6	**45.5**	29.5	0.33	**0.40**
50	34.2	**30.2**	11.9	0.29	**0.37**
Full	24.7	**21.2**	—	0.23	**0.29**

FID: lower is closer to real. Recall: share of real images the generator covers. Both against an equal-size real sample.

LoRA is closer to the real data

Lower FID from 10 per class up, and higher recall at every budget.

…but SD class prompts train better

For pretrained models, plain prompts gave higher accuracy than LoRA at all 5 budgets, for both classifiers.

An early sign for RQ3; Stage 4 tests it with more generators.

What Stage 1 tells us so far

1

Scarcity decides the value

With pretrained models, synthetic gains are largest at 5 images per class (+10 to +21 points over real only) and about one point or less on the full set.

2

Add it to augmentation, don't swap it in

SD + RandAugment beats RandAugment up to 50 per class; SD alone loses to RandAugment from scratch.

3

Pretraining changes the answer

Pretrained models use synthetic images well; scratch models gain more from RandAugment.

4

The Transformer gains more, then less

ViT-B/16 gains the most at 5 per class but falls behind RandAugment earlier as data grows.

Thank you